

# Econ 30065: Microeconomics

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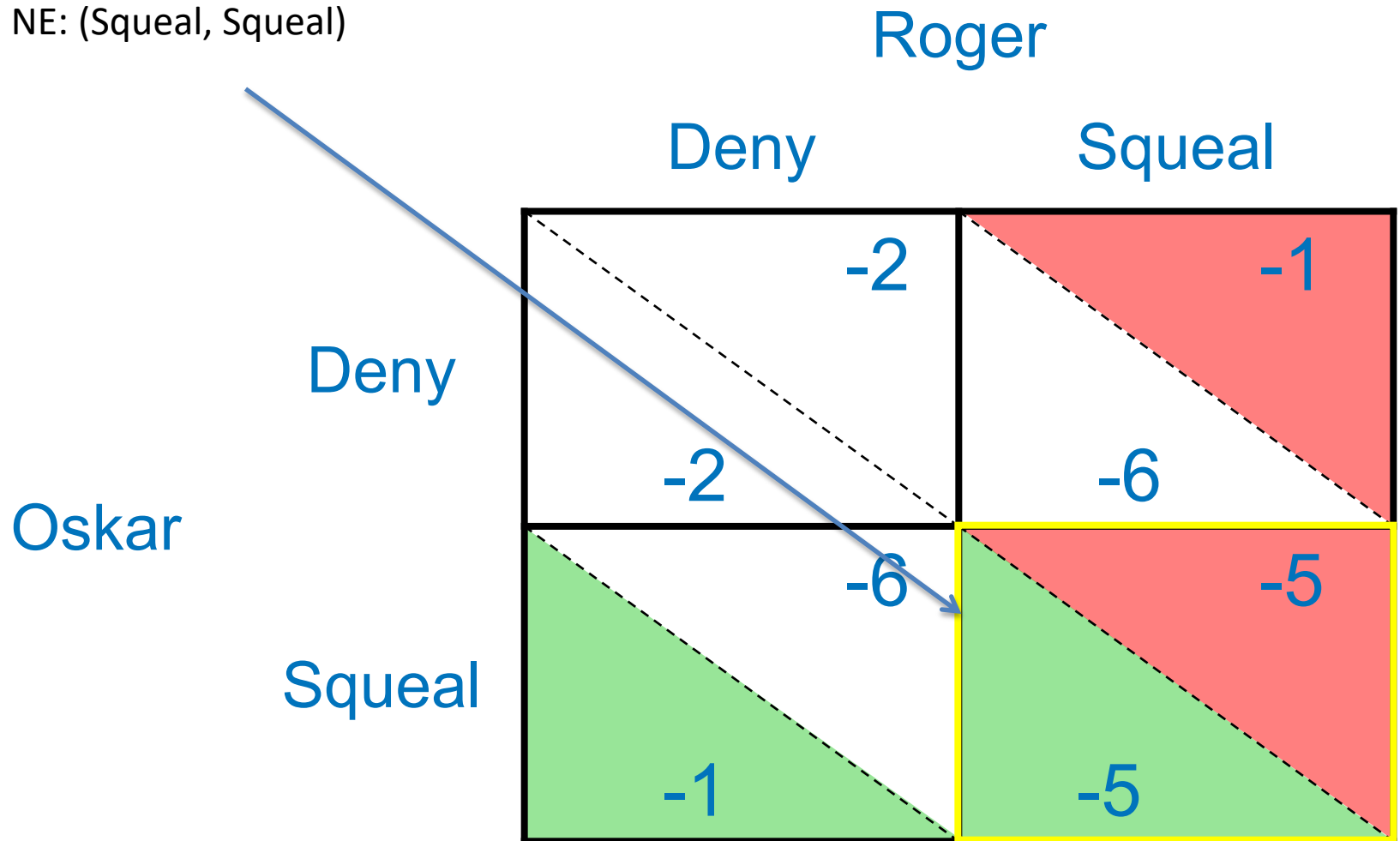
Game Theory: One-stage games

# Nash Equilibrium in One-Stage Games

- Concept created by mathematician John Nash, published in 1950, awarded Nobel Prize
- Has become one of the most central and important concepts in microeconomics
- In a ***Nash equilibrium***, the action played by each individual is a best response to the actions played by everyone else
  - Everyone correctly anticipates what everyone else will do and then chooses the best available alternative
  - Combination of action in a Nash equilibrium is stable (people are doing their best, given the action of the opponents)
  - In other words, once players have seen action chosen by others, no one wants to unilaterally deviate from what he chose.
- A game can have 1, 2 or more NE in pure strategies, or also NO Equilibria in pure strategies
- In practice:
  1. for each action of the opponent, calculate the best responses for each player
  2. Find the profiles of actions where both players are playing best responses

# Nash Equilibrium in the Prisoners' Dilemma

NE: (Squeal, Squeal)



# Battle of the Sexes: many equilibria

|       |                 | Tony         |                 |
|-------|-----------------|--------------|-----------------|
|       |                 | Action movie | Romantic comedy |
| Maria | Action movie    | 5<br>2       | -1<br>-1        |
|       | Romantic comedy | 1<br>1       | 2<br>5          |

# Nash Equilibria for the Battle of the Sexes

|       |                 | Tony         |                 |
|-------|-----------------|--------------|-----------------|
|       |                 | Action movie | Romantic comedy |
| Maria | Action movie    | 5<br>2       | -1<br>-1        |
|       | Romantic comedy | 1<br>1       | 2<br>5          |

# No Nash Equilibria

|        |                           | Vizzini                                         |                                                 |
|--------|---------------------------|-------------------------------------------------|-------------------------------------------------|
|        |                           | Drink from<br>left goblet                       | Drink from<br>right goblet                      |
| Wesley | Poison in<br>left goblet  | <div><div></div><div>1</div><div>-1</div></div> | <div><div></div><div>-1</div><div>1</div></div> |
|        | Poison in<br>right goblet | <div><div></div><div>-1</div><div>1</div></div> | <div><div></div><div>1</div><div>-1</div></div> |

# Iterative deletion of dominated actions and Nash Equilibrium

- If a player has a dominant strategy, he will also play it in Nash Equilibrium!
- Dominated actions cannot be part of an equilibrium
- If the first 2 solution procedure lead to 1 outcome, that is a Nash Equilibrium

# Nash equilibrium and welfare

Sometimes agents are both worse off in terms of payoff when they play the Nash Equilibrium than if they played another pair of actions (but still NE is the only stable outcome)

- That happens in the prisoner dilemma

# Nash Equilibria in Games with Finely Divisible Choices

- Concept of Nash equilibrium also applies to strategic decisions that involve finely divisible quantities (infinite number of actions)
- Determine each player's best response function
- A ***best response function*** shows the relationship between one player's best response and other players' choices
- A pair of choices is a Nash equilibrium if they satisfy both best response functions simultaneously

# Example: Free Riding in Groups

- Scott and Liz need to write a research report together. They have 24 hours to do the research. Their promotion depends on how good is this report. Each conducts research simultaneously and independently
- Their choice is how much time to spend on the report
- They can't observe each other's choice.
- **Free-riding problem:** each player would prefer to spend less time on the report if the other is spending enough time on it
  - the best response functions will be downward sloping. If Scott (Liz) knows Liz (Scott) will work hard, he (she) will work less.
- Intersection (N.E.) – the only point at which each player's action is the best response to what the other player is doing.

# Mathematically

Assume  $U^i(X,Y)=40(X+Y)-(X+Y)^2$  where  $i=S,L$  and  $X$  is the number of hours chosen by Scott,  $Y$  is the number of hours chosen by Liz, (and  $X,Y$  must be positive and smaller than 24). Moreover the cost of time effort is  $X^2/2$  and  $Y^2/2$

How many hours each player decide to work?

# Solution

1) Find Best Response functions:

1) Scott:  $40 - 2(X + Y) = X \rightarrow X = (40 - 2Y)/3$

2) Liz:  $40 - 2(X + Y) = Y \rightarrow Y = (40 - 2X)/3$

2) Find the point where they cross

1)  $X = (40 - 2(40 - 2X)/3)/3$

$\rightarrow X = 8, Y = 8$ : Nash Equilibrium

# Nash Equilibrium

